

GT2101 Debugging Reference

Facts-only technical reference for the Gale GT2101 control tower: board functions, signal chain, and pin-to-pin test points. No history, no provenance, no attribution — bench use only.

1. System overview

Five PCBs in the control tower (Board 1 through Board 5), connected by a single flexible backplane film ("the flexicon"). One motor PCB inside the motor pod, wired to the tower via the flexicon.

Signal chain, demand side (speed-setting): Helipot (3-wire pot) -> Board 3 (XR2207 VCO) -> F VAR output -> Board 5 (black switch, VAR/FIX select) -> Board 4 -> Board 5 pad 4 (selected x40 demand) -> feeds both the servo and the display.

Signal chain, feedback side (actual speed): Motor optical encoder -> motor PCB (LM339 tachometer comparator) -> TACH -> Board 4 -> compared against demand -> drive correction back to motor PCB -> motor windings.

Display chain: Board 2 divides a 1.048711 MHz crystal (arriving from Board 4) with an MC14521 24-stage divider, producing a gating signal. A 4016 switch selects between two divider outputs for the gate rate, and separately selects F DISPLAY between demand (F VAR/FIX) and actual (INV TACH). Board 1 counts F DISPLAY pulses during the gate window and drives the LED digits.

2. Board 1 — Display logic

Counts F DISPLAY pulses over the gate window and drives the LED digits.

- Pad 2 — BLANK
- Pad 3 — F DISPLAY (in, from Board 2 pad 8)
- Pad 4 — GATE (in, from Board 2 pad 10)
- Pad 6 — RESET (out, to Board 2 pad 13, drives the MC14521's R pin; listen-only from Board 1's side — never drive this pin, it re-zeroes the tower's display timebase)
- Pad 7 — F REF (in, from Board 4 pad 9, the MC14520 divide-by-4 output)
- Pad 12 — ground

Internal logic: 4001 gate D = NOR(4013 Q2, GATE) feeds 4013 R1/R2, so GATE parks the display's internal divider during the count window and releases it when the window closes. Released, F REF walks the 4013 through four states — state B = latch (4011 gate C = NAND(Q1, NOT-Q2)), state D = reset (4001 gate C = NOR(NOT-Q1, NOT-Q2)).

Count arithmetic: 1332 Hz x 0.25 s window = 333 counts -> displays "33.3".

3. Board 2 — Timebase and mode switching

- Pad 2 — crystal signal in (1.048711 MHz, from Board 4), measures 5.2 V DC on a meter (it's a square wave, meter reads half the 10.7 V rail)
- Pad 3, Pad 4 — STILL / TURNING inputs (from Board 3's LM3900, via the flexicon)

- Pad 6 — logic node behind a 10k and a clamp diode; Board 1's end has a pull-up to +10 V; rests high, pulses low (unconfirmed by meter as of last audit)
- Pad 7 — F VAR/FIX demand in
- Pad 8 — F DISPLAY out (to Board 1 pad 3)
- Pad 9 — INV TACH in
- Pad 10 — GATE out (to Board 1 pad 4)
- Pad 11 — MC14521 Q19 output, measures 3.7 V (not 50% duty — correct for a divider with a NAND reset)
- Pad 13 — RESET in (from Board 1 pad 6), drives the MC14521's R pin; rests low, pulsed high
- Pad 14 — ground reference for this board

ICs: MC14521 (24-stage divider off the crystal; Q21 = 0.4999 Hz, Q19 = 2.000 Hz) and a 4016 two-way switch (selects which divider output becomes GATE, and separately selects F DISPLAY between demand and actual, both switched together by the same STILL/TURNING control bit from Board 3 pads 3/4).

Known fault signature: pad 2 reads 5.2 V (crystal present) and pad 11 reads 3.7 V (first divider stage running) but pad 10 reads near 0 V — the MC14521 or the 4016 channel feeding pad 10 is faulty; connector pins alone can't distinguish which.

4. Board 3 — Speed demand (Helipot / XR2207 VCO)

Takes the Helipot's wiper voltage and converts it to a frequency (F VAR) for the servo and display.

Helipot wiring (3-wire pot, legend R 1K, L .25, code 7603):

- RED post — pot's CCW leg — board 3 GND
- ORANGE post — pot's wiper (S) — XR2207 pin 6 (timing resistor terminal) AND, via a capacitor, LM308 pin 3 (+ input) — this post does two jobs
- YELLOW post — pot's CW leg — board 3 negative rail (nominally -10 V)

XR2207 (14-pin VCO):

- Pin 6 — timing resistor terminal, fed from the ORANGE post through a resistor (pins 4, 5, 7 are NC, so pin 6 is the only timing resistor — open circuit here means zero timing current)
- Pin 13 — SQUARE OUT (second pin from the left, top row); 10.66 kOhm pull-up, 10.16 kOhm series resistor to a 4011 inverter (pins 5+6 in, pin 4 out) -> pad 5
- Pin 11 — BIAS (not SQUARE OUT — do not confuse)
- Pin 12 — negative supply

LM308 (fed by the same ORANGE post via a capacitor, AC-coupled):

- Pin 3 (+ in) — from ORANGE post
- Pin 6 output — through a resistor to LM3900 "1+ IN" and "2- IN" (same wire, both inputs)

LM3900 (quad op-amp, two independent pairs):

- Amps 1 and 2 — window comparator off the LM308; "2 OUT" -> pad 2, "1 OUT" -> pad 3 (STILL/TURNING lines to Board 2)
- Amps 3 and 4 — separate pair, senses pad 7 net (IN 3-, pin 8) — drives 4011 -> 4013 -> BC214 -> J113, i.e. motor mute / touch-pulse latch / 0V-at-rest. This pair is independent of the ORANGE post and

the window comparator.

Backplane routing (confirmed on backplane.pdf): pad 5 (VAR, F VAR output) runs straight to Board 5 row 5 pad 4. It does NOT pass through Board 2, and there is no x40 multiplication on this leg — it leaves Board 3 already scaled at 40 Hz per rpm ($3996 \text{ Hz} = 99.9 \text{ rpm} \times 40$; $1332 \text{ Hz FIX} = 33.3 \text{ rpm} \times 40$).

Test points:

- Pad 5 to pin 9 (ground), black switch released (VAR position): should swing rail-to-rail, 0 V low to 10 V high. If stuck around 7.5 V, the black switch is pressed (FIX) — not a fault.
- A multimeter cannot resolve F VAR directly — at ~1333 Hz it reads a drifting average indistinguishable from a dead line. Use a scope, or hold the probe still and watch for movement.

Transistor leg identification (when fitted as a salvage part, not from a datasheet pinout — TO-92 pinouts mirror depending on which face you read): identify by function, not position — base = control leg (gets the drive resistor), emitter = return leg (to ground), collector = working leg (to the circuit). Confirm by testing for off->1, on->0 switching, not by counting legs.

5. Board 4 — Servo / tacho front end

- Pad 2 — crystal distribution point (1.048711 MHz out to Board 2 pad 2)
- Pad 3 — selected x40 demand in (same net as Board 5 pad 3 and Board 2 pad 7)
- Pad 8 — common ground reference point for cross-board continuity checks
- Pad 9 — F REF out (MC14520 divide-by-4 output) -> Board 1 pad 7
- Row 4, pad 5 — TACH in point (listen-only tap point for an external monitor)

Crystal: 1.048711 MHz, confirmed running. ~600 ppr (counts per revolution) encoder resolution figure appears on this board's documentation.

Known-good reading: Board 4 pad 2 and Board 2 pad 2 should read identically (both ~5.2 V) when ground is common between the two boards. If they read differently despite both beeping continuous on a continuity test, ground is NOT actually common — stop and re-establish ground before trusting any other voltage on either board.

6. Board 5 — Power supply and mode switching

Two independent rails, made differently:

- Positive rail: through a 3-terminal device (location: under the heat-sink bracket, corner of board — NOT confirmed to be the same part position on every board; verify by eye for each physical board). Nominal $\approx +11.2$ to $+11.9$ V. Pad 2 is the positive rail test point.
- Negative rail: zener + discrete PNP pass transistor (PNP hfe=283 per schematic). Nominal ≈ -10 to -11 V. Pad 8 is the negative rail test point. PNP pinout viewed from the top, flat edge: pins are B, C, E in that order.

Common ground: pad 9.

Other components on this board: triac 2N6342 (TO-220, free-standing, middle of board beside the diac and a white trimmer — NOT the part under the heat-sink bracket), diac 1N5761A (2 legs), bridge rectifier 10DB1A (+

corner marked), 2x 100 uF 25 V blue axial capacitors, 560 ohm resistor (regulator section).

Row 5 / pad map (backplane):

- Pad 1 — LED GREEN
- Pad 2 — +10 V (nominal; measures ~11.7 V in practice)
- Pad 3 — black switch common (VAR/FIX selector, shared net with Board 4 pad 3 and Board 2 pad 7 — this is the selected x40 demand signal)
- Pad 4 — F VAR in (direct from Board 3 pad 5, no intermediate board)
- Pad 7 — fixed 1332 Hz (FIX reference)
- Pad 8 — negative rail (should read approximately –10 to –11 V to ground at pad 9; a reading near +1.6 to +1.7 V indicates the negative rail is not being generated — check pad 2 for +11.7 V first to rule out a blown fuse as the cause)
- Pad 9 — ground reference for this board

Standard reading table, positive rail healthy:

- Pad 2 to pad 9: +11.7 V
- Bridge + terminal to pad 9: approximately 1.1 MOhm (both probe directions)

Fault isolation note: a dead negative rail (pad 8 near +1.6 V instead of –10 V) will starve both Board 3's XR2207 (no negative supply for the VCO) and Board 3's LM3900 (no negative supply for the STILL/TURNING window comparator) simultaneously — both faults can share one root cause.

Black switch (VAR/FIX): a passive mechanical switch physically located in the tower base casting, not on Board 5 itself — Board 5 pads 3/4/7 are only the route out to it. Two switch positions select between F VAR (pad 4, variable) and the fixed 1332 Hz reference (pad 7). Also grounds the green LED in one position. EAO type 01-272 pushbutton, contacts rated 300 Vac 5/250, lamp rating 60 V / 1.2 W.

7. Motor PCB (inside motor pod)

Single analogue input net: SPEED IN, arriving from the control tower. This is one analogue amplitude command — the motor PCB itself handles commutation. No three-phase PWM is needed from outside; one analogue voltage level is sufficient.

SPEED IN reaches each of three phase chains through 47 kOhm + 47 kOhm (two resistors in series), with a capacitor to ground at the junction between them. Each phase chain has an N-channel JFET (type J113, substitution for an original E113) shunting that junction to ground, gated by that phase's own comparator output.

LM339 (quad comparator) — all four sections in use:

- Position sensor GREEN: inputs pins 8, 9 -> output pin 14
- Position sensor YELLOW: inputs pins 6, 7 -> output pin 1
- Position sensor VIOLET: inputs pins 10, 11 -> output pin 13
- Each of the above three has 470 kOhm feedback from output to the + input (positive feedback = hysteresis, for clean squaring of the sensor signal)
- Fourth section: inputs pins 4, 5 -> output pin 2 — this is the TACH comparator. Its input is a wire labelled BROWN. The tachometer's open-collector output pulls to ground through 4.7 kOhm; the chip's own

negative rail is -10 V , which is why TACH swings between 0 V and -10 V .

LM324 (quad op-amp) — three sections in use, positioned after the JFET gating network:

- Section: inputs 6, 5 -> output 7
- Section: inputs 2, 3 -> output 1
- Section: inputs 13, 12 -> output 14
- Each drives an output pair through a 560 ohm resistor to the motor windings
- Fourth section (inputs 9, 10 -> output 8) is drawn NC — unused spare on this board

Winding pairs (three), each pair color-coded: green/white, red/white, black/white — these terminate at the LM324 output stages.

Rails on this board: output Darlington transistors run on $\pm 15\text{ V}$. Both the LM339 and LM324 run on a -10 V negative rail (shared with the tower's negative rail). Sensor bias bus reaches that negative rail through 180 kOhm.

TACH signal: the wire labelled BROWN feeds the LM339's fourth (tacho) comparator section directly — it is NOT derived from any of the three winding pairs.

8. Flexicon / backplane — key cross-references

The backplane is a single flexible film connecting all five boards. Always solder to the brass staples, never directly to a pad.

Confirmed single-net groupings (same electrical net, multiple pad numbers across boards):

- Board 5 pad 3 = Board 4 pad 3 = Board 2 pad 7 — one net, the selected x40 demand signal (output of the VAR/FIX switch, feeds both servo and display)
- Board 3 pad 5 (VAR output) -> Board 5 row 5 pad 4 directly — no intermediate board, no multiplication on this leg

Ground continuity reference chain (prove common before trusting any voltage reading): Board 5 pad 9 -> Board 4 pad 8 -> Board 3 pad 9 -> Board 2 pad 14 -> Board 1 pad 12.

Scaling reference: F VAR leaves Board 3 already at 40 Hz per rpm of platter speed.

- 33.3 rpm -> 1332 Hz (FIX reference value)
- 99.9 rpm -> 3996 Hz

9. Quick fault-to-pin lookup

Symptom	Check first
Display dead / reads garbage	Board 2 pad 2 (crystal in, expect 5.2 V) then pad 11 (expect 3.7 V) then pad 10 (gate out — near 0 V here with the first two present means MC14521 or 4016 fault)
Speed control (knob) has no effect	Board 3 ORANGE post (Helipot wiper) for a smooth sweep; then XR2207 pin 6 for timing current; then pin 13 for SQUARE OUT

Symptom	Check first
	swing
VAR mode won't engage / sticks in FIX	Check the base casting's black switch (VAR/FIX) and its two contact pairs — one pair is lamp, one is contacts, not yet distinguished by position
Everything analogue dead at once (VCO, STILL/TURNING, servo)	Board 5 pad 8 to pad 9 — if near +1.6 V instead of −10 V, the negative rail is down; check pad 2 first (+11.7 V) to rule out a blown fuse
No drive to motor at all	Confirm SPEED IN is present at the motor PCB; then check each phase's JFET gate (comparator output) is toggling
TACH reads nothing	LM339 fourth section, pins 4/5 in, pin 2 out, on the motor PCB; confirm the BROWN wire is intact to pin 4/5
Two boards "beep continuous" on continuity but read different voltages	Ground is not actually common between them — stop, re-establish a verified common ground before taking any further reading

Facts only — pin numbers, nets, components and measured/predicted values as documented and bench-verified. No restoration history, provenance or attribution included.

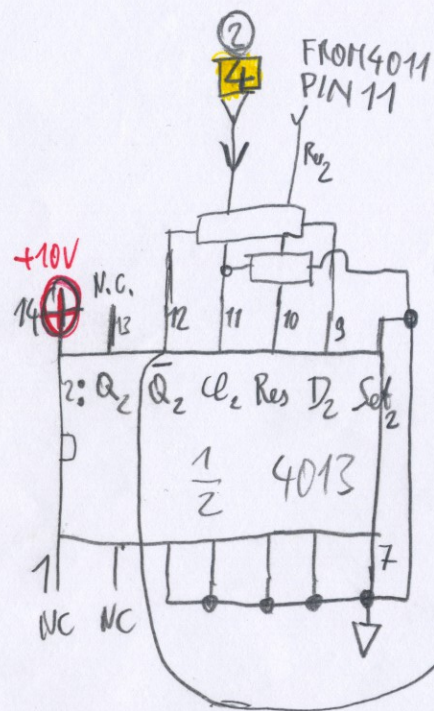
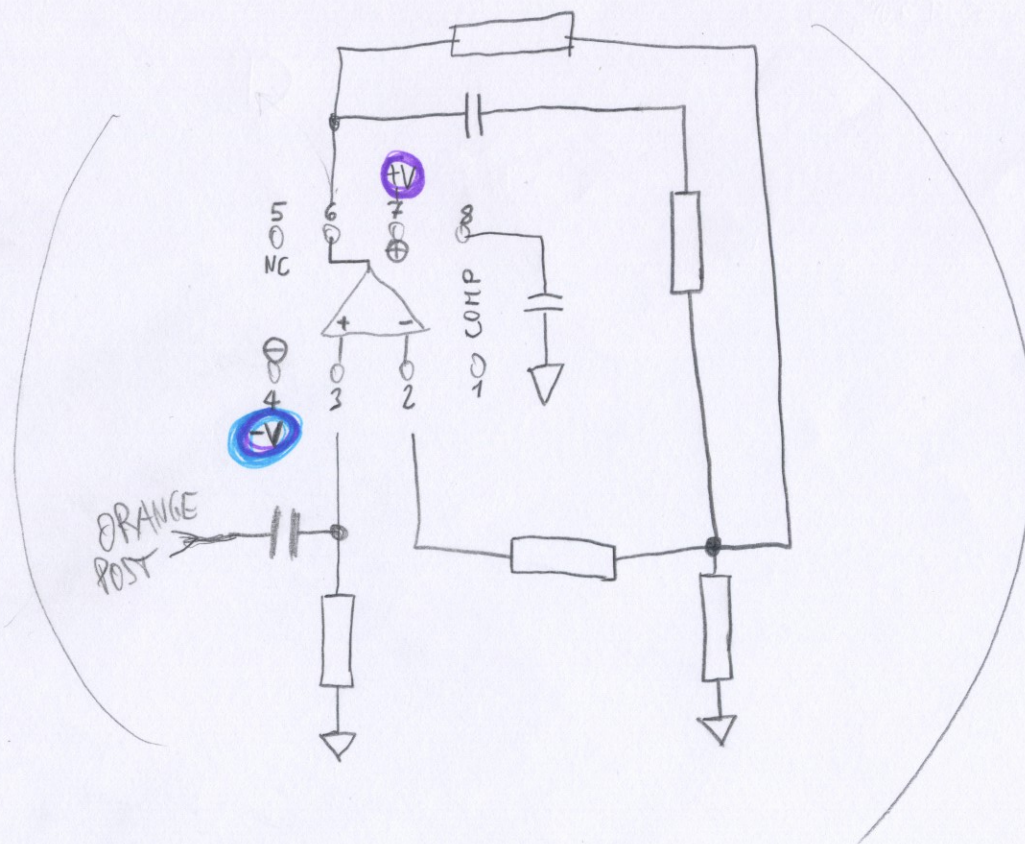
③ A

LM308

(A-D)

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PRELIMINARY



Replaced by 1N4148

Defective Replaced by J113 (RS-components)

BC214
PNP

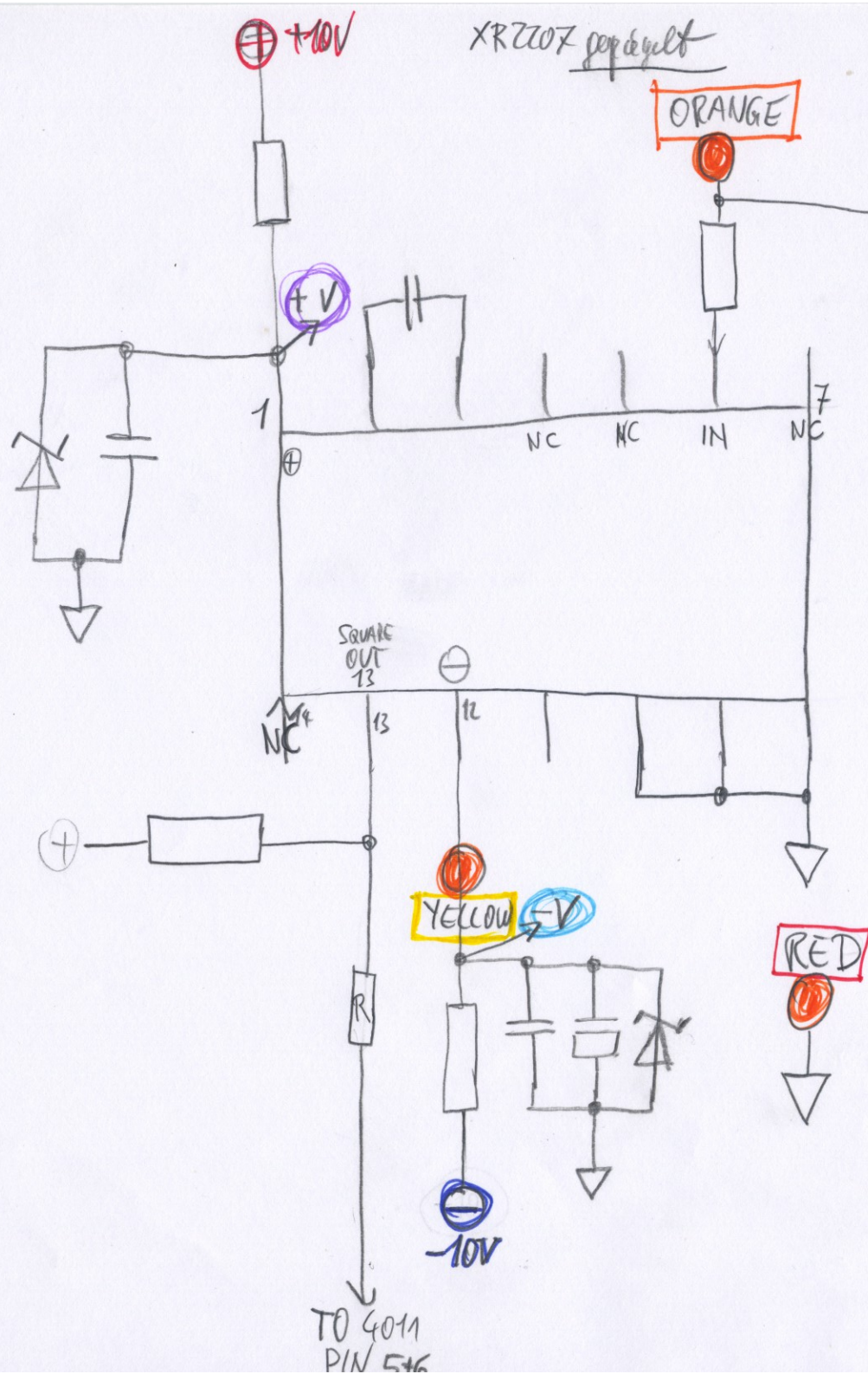
E113

4013

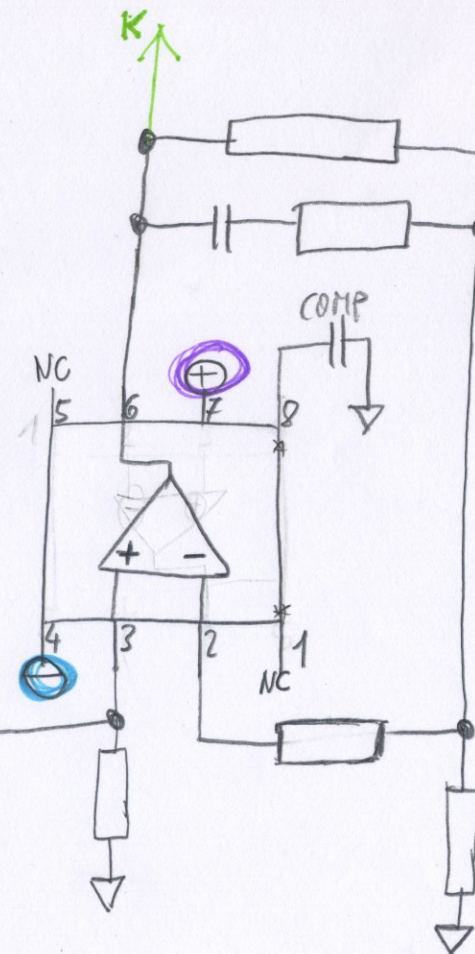
-10V

(3) B

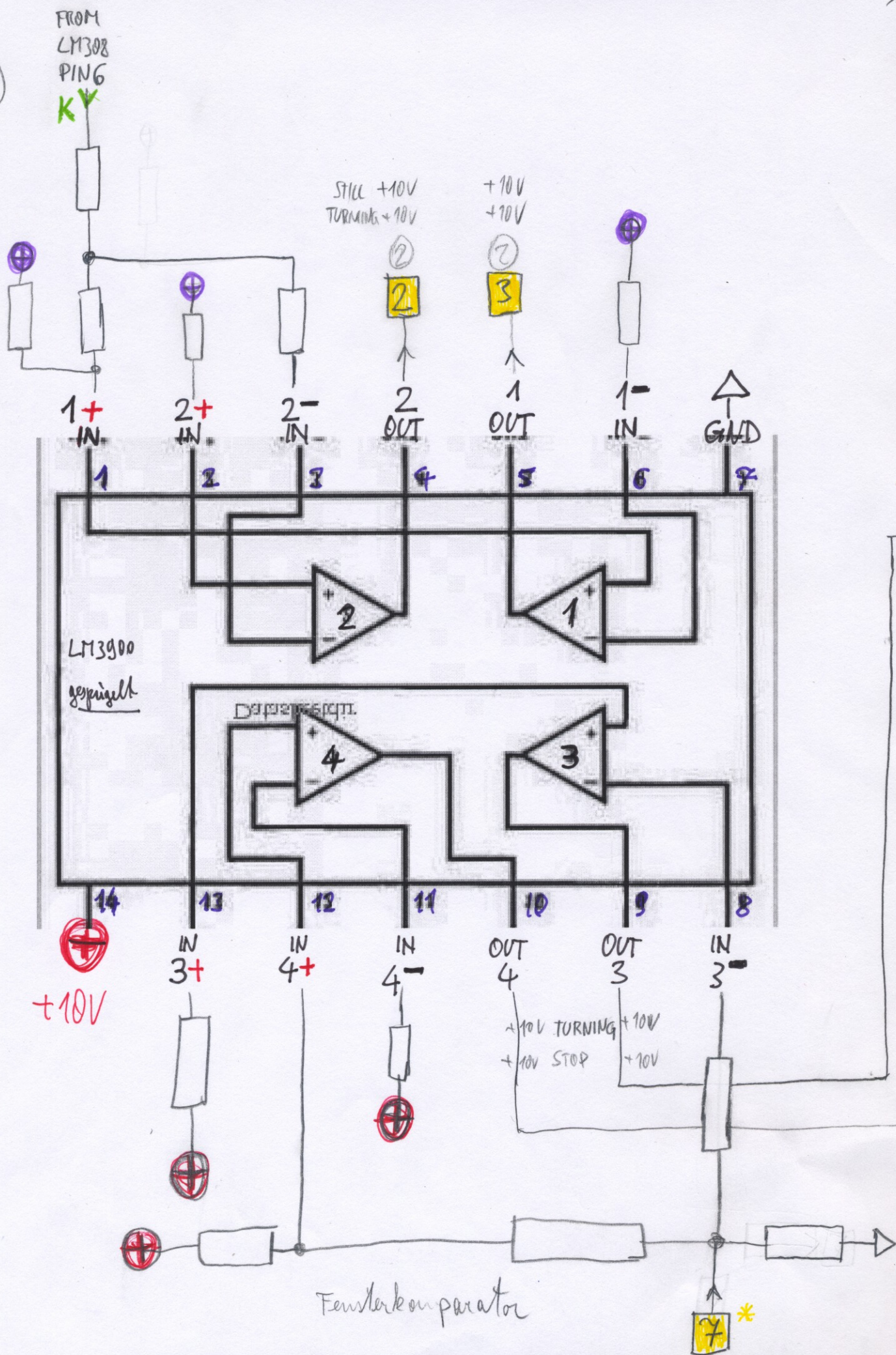
(A-D)



LM308 gepüßelt

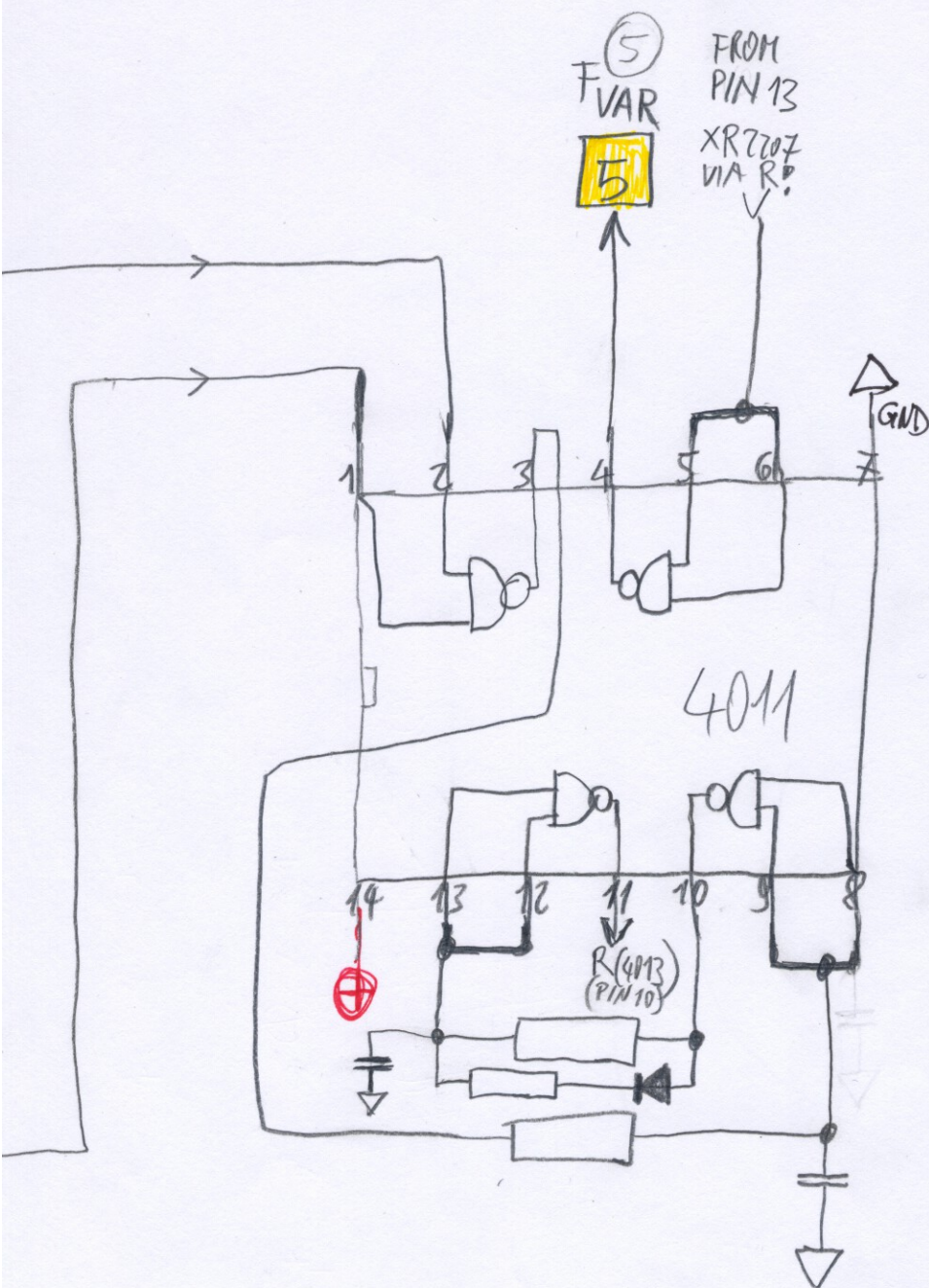


③ C
(A-D)



LM13900 MIRRORED!

③ D
(A-D)



4011 MIRRORED!

1A GND 8

+10V 1

START/STOP
LOW PULSE
(FROM PCB ②)

F_{DISPLAY}
(FROM PCB ②)

+10V

LED
V_{GN}

5

GREEN
LED
TO PCB ⑤

4011

5 4
6 NOT USED
M.C.

4
1Hz
From
PCB ②

TO 4553
PIN 10
"LATCH ENABLE"

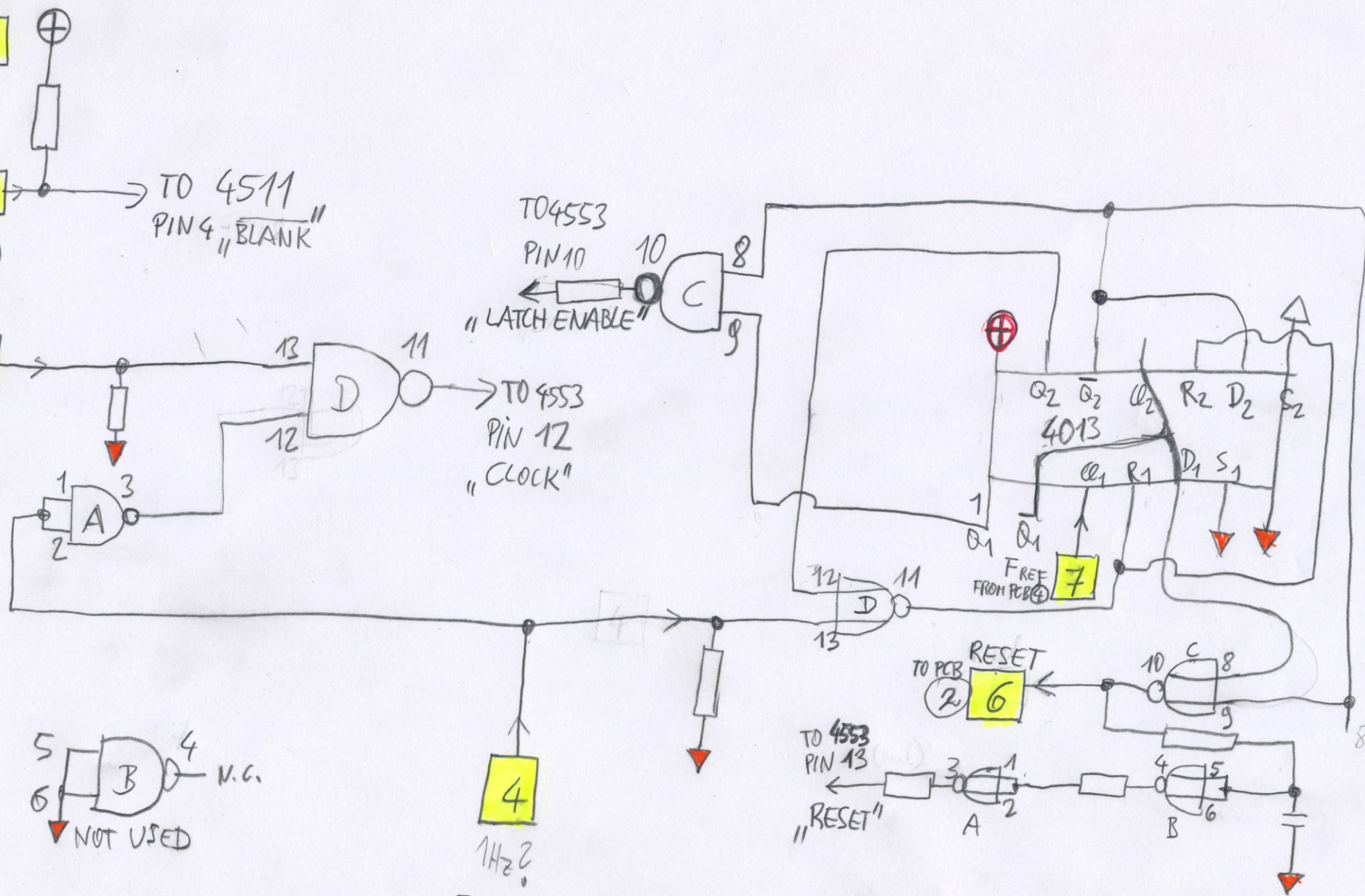
TO 4553
PIN 12
"CLOCK"

TO PCB
RESET
② 6

TO 4553
PIN 13
"RESET"

4001

4013



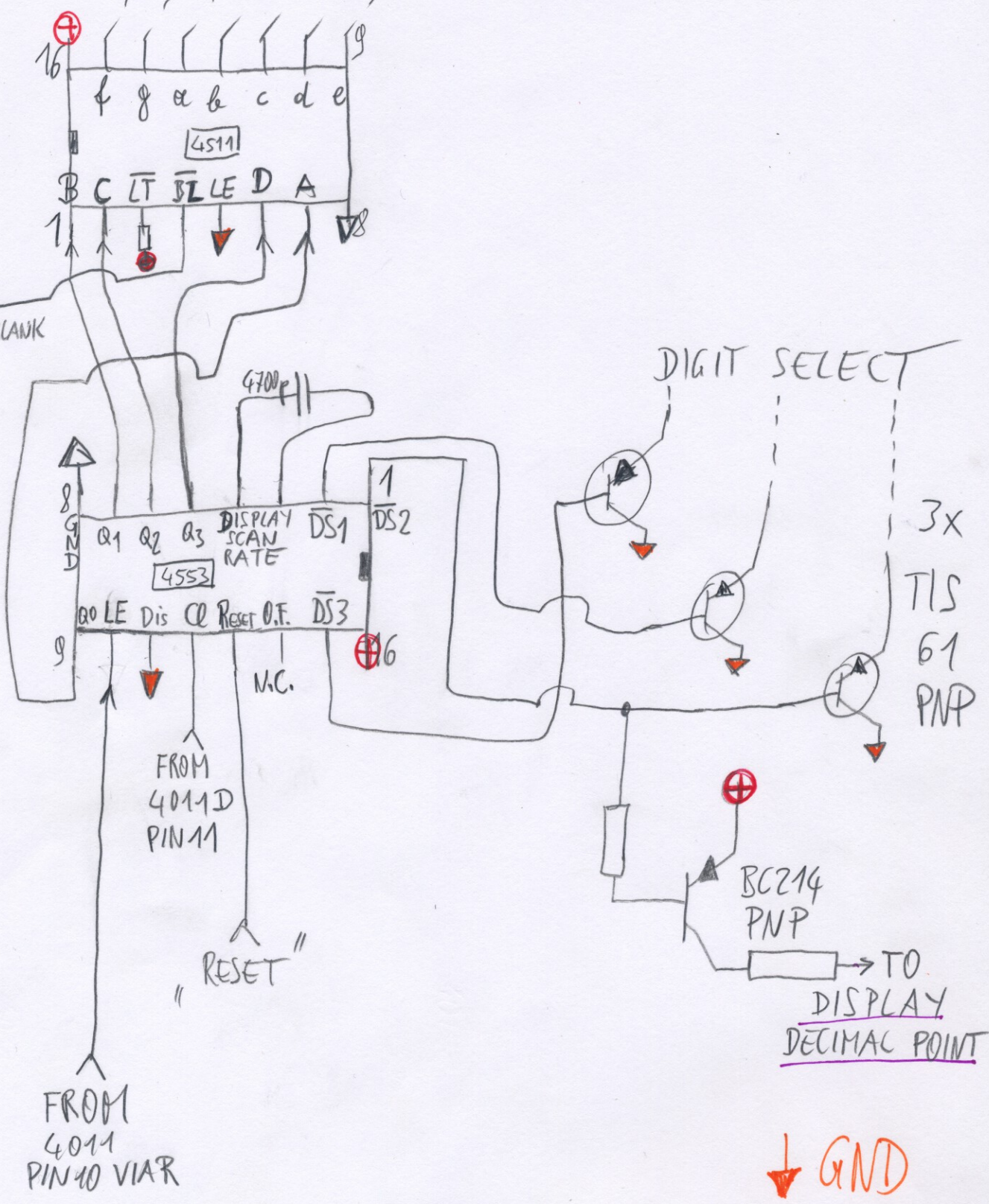
1B

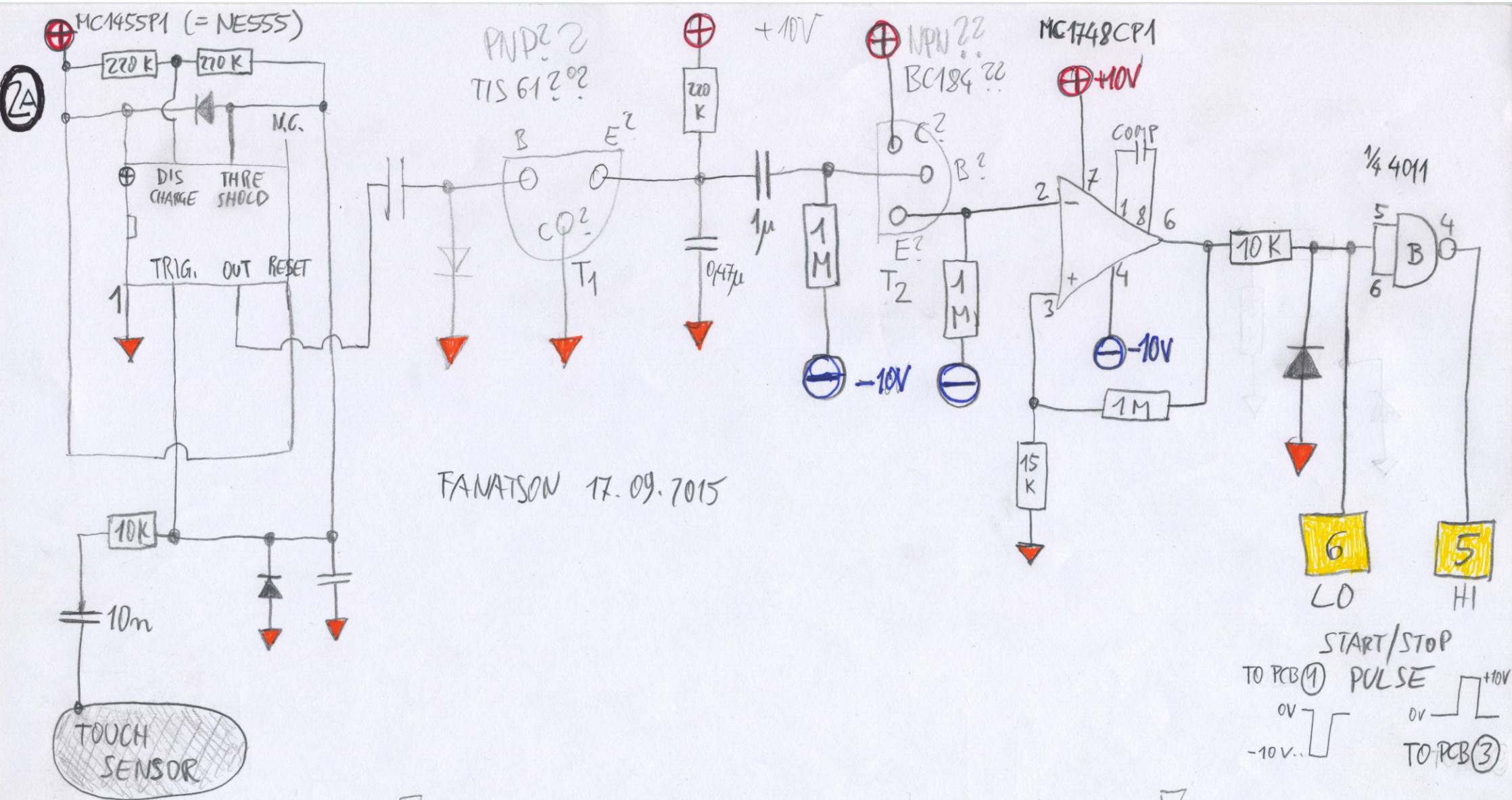
TO LCD SEGMENTS'

4511

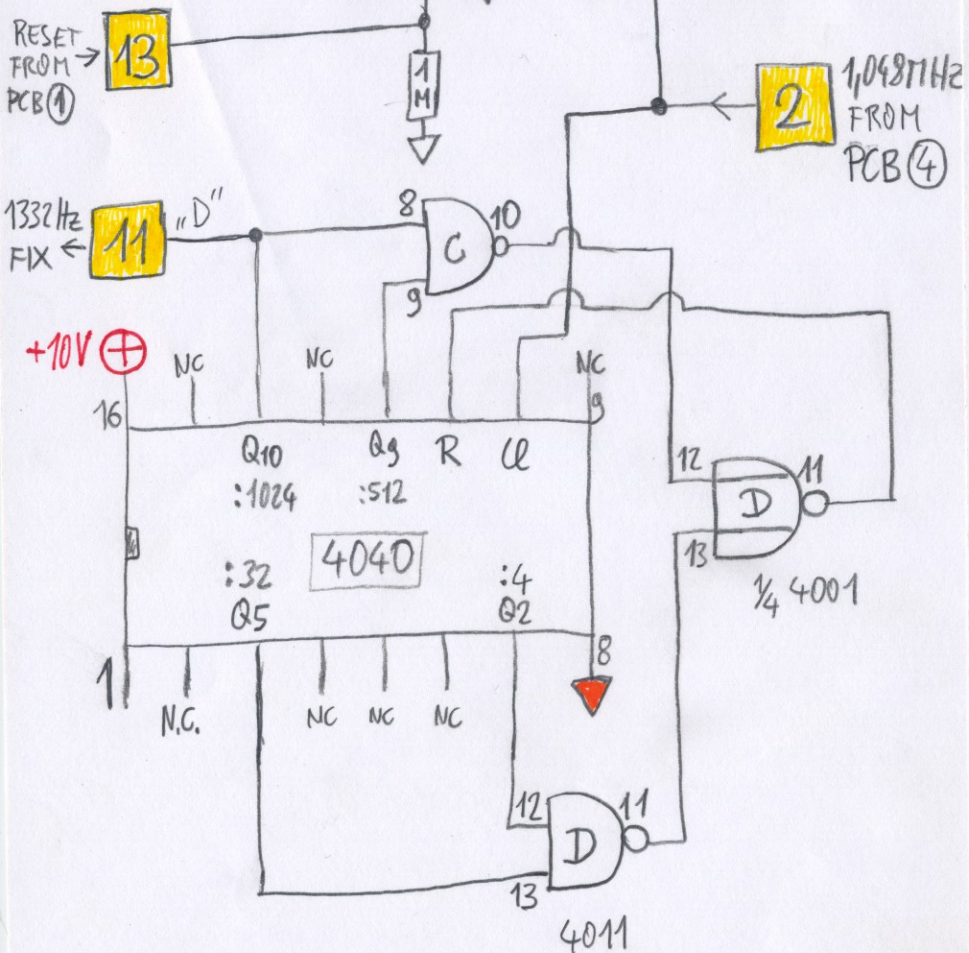
FROM 2
LOW PULSE → BLANK

4553

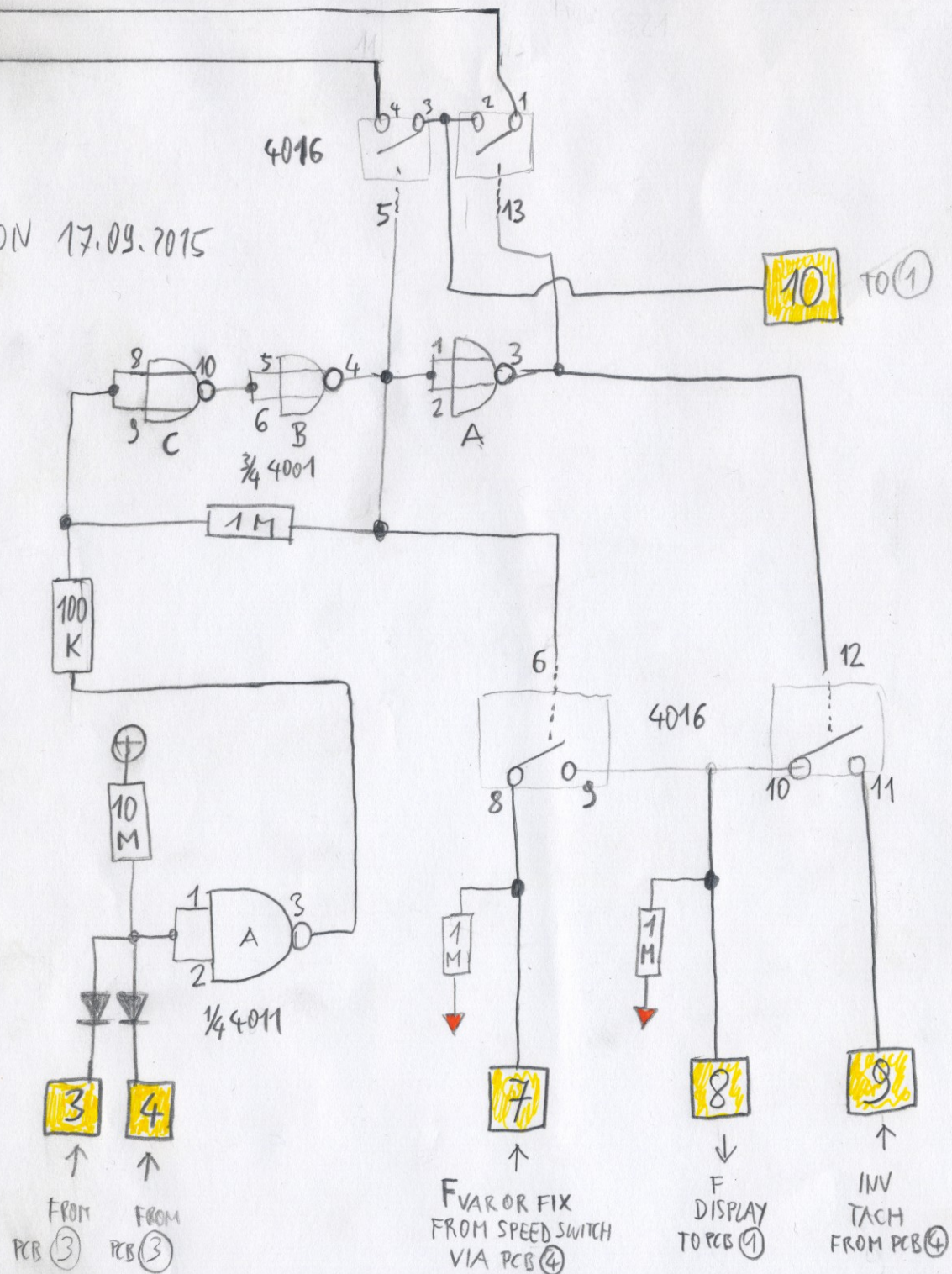


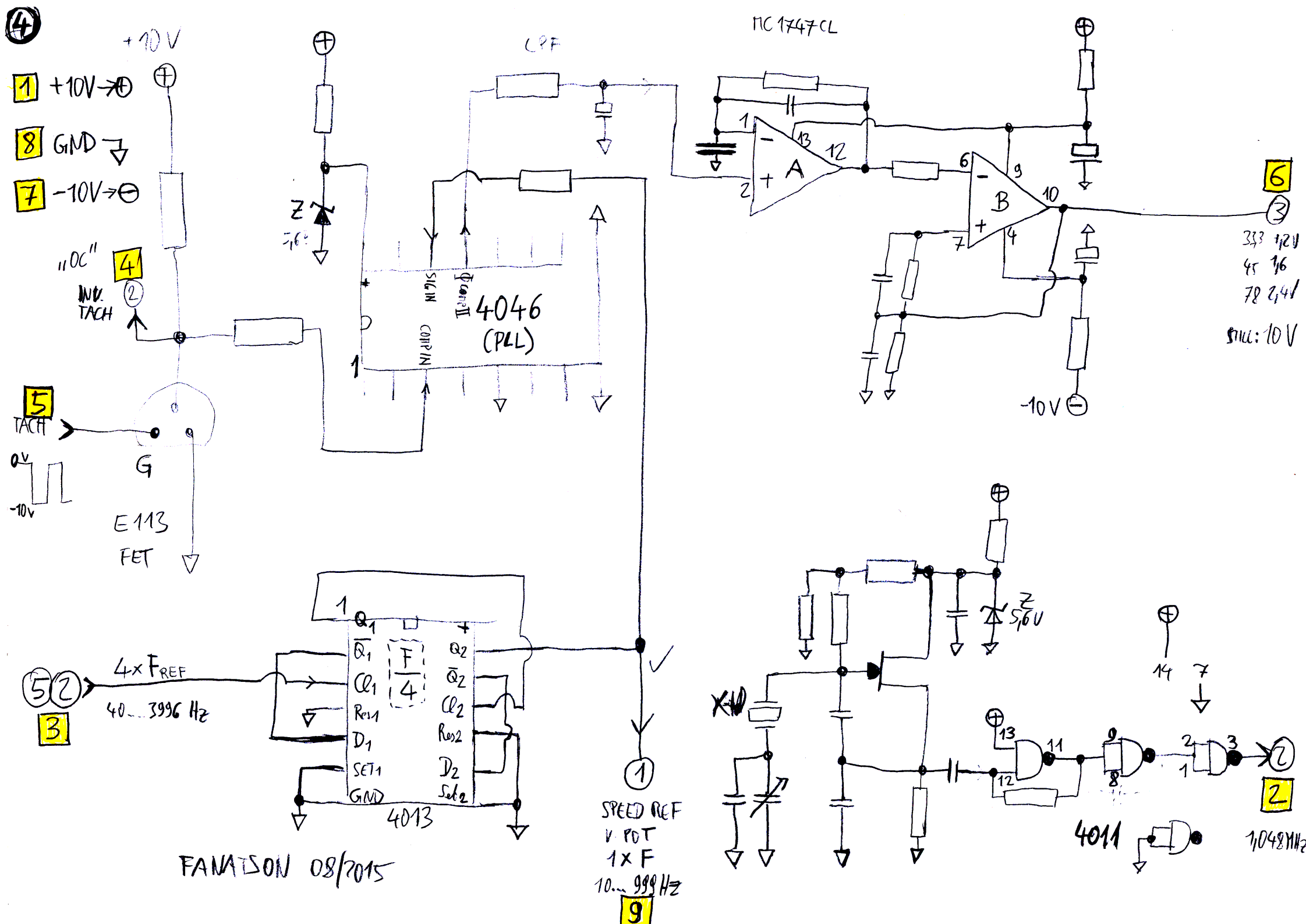


GND 14

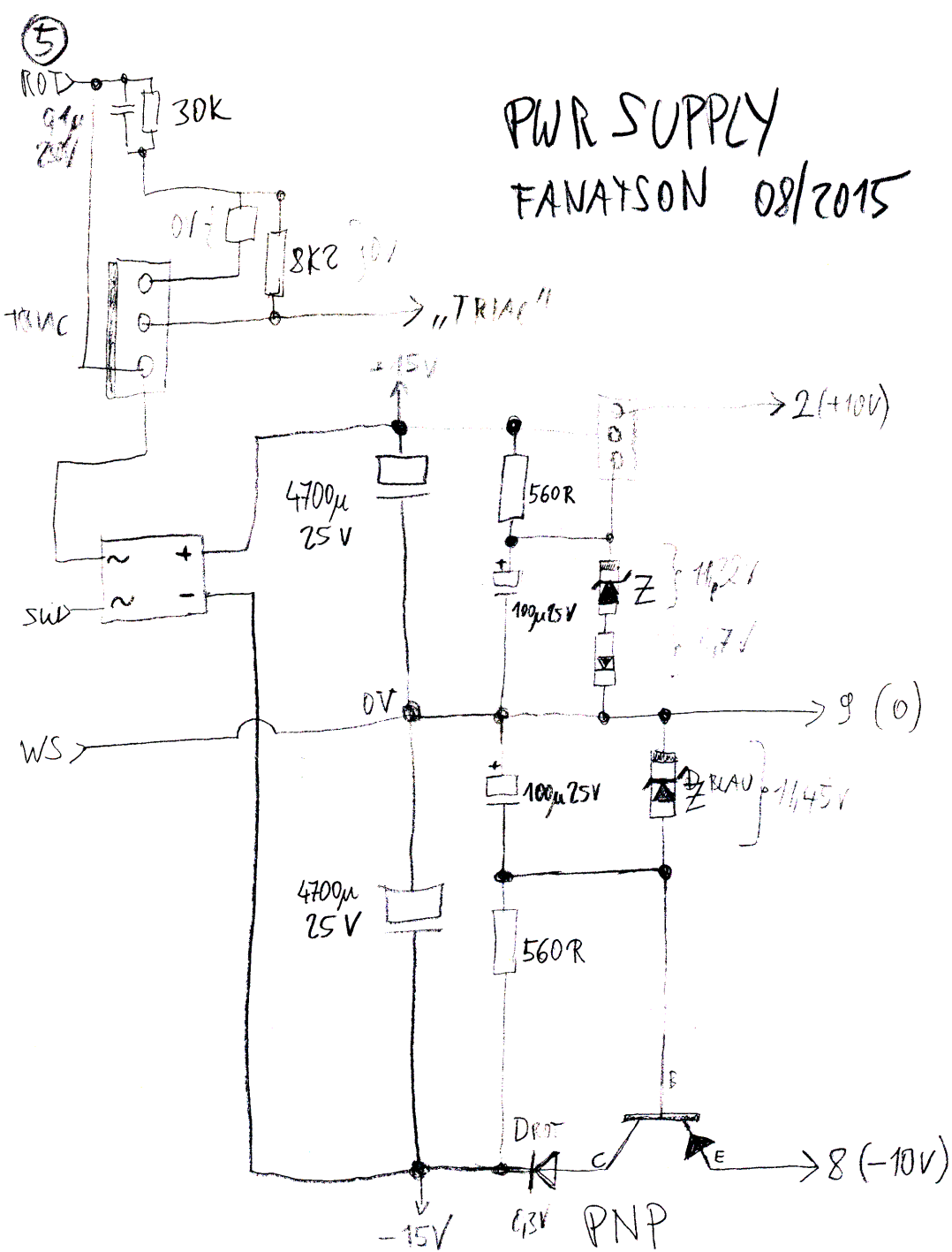


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E_{BV}
 GE^2
 PNP
 $h_{fe} = 283$
 $V_{BE} = 0.69 V$


 TOP VIEW.

LM339

SPEED IN 1/2"

JFET NCH
E113
J113

LM324

BD675A PNP DARLINGTON
BD676A PNP DARLINGTON

BROWN (N)

(M)

TACH

MOTOR PCB

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